Enhancing Real-Time Point-Cloud Streaming through In-Network Computing

Master's Degree in Engineering in Computer Science

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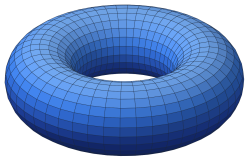
Academic Year 2025 / 2026



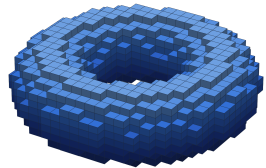
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From XR to Point Cloud Streaming

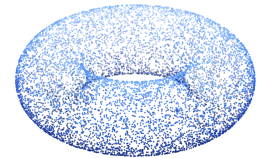
1 Introduction



Polygonal mesh



Voxel grid



Point Cloud

Context and Motivation

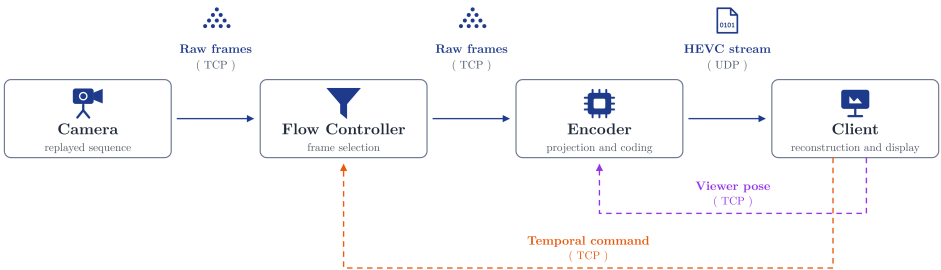
- **Extended Reality**: virtual, augmented, mixed
- One timing budget, sensing to display [4, 14]
- **Six degrees of freedom**, chosen after capture

The 8iVFB Dataset

- **Loot**, from the **8i Voxelized Full Bodies** [2]
- 300 frames of about 794 thousand samples
- Quantised on a lattice of 1.820 mm [15]

Architecture and Interfaces

1 Introduction



Four coarse application blocks are joined by a transport that carries bytes and inspects none of them [8]. Projection, coding and both reverse loops sit inside the **Encoder**, which therefore sets the pace of the chain [5, 12].



Limits of Real-Time Volumetric Workflows

1 Introduction

Throughput

- 3 Gbit / s of raw samples at nominal cadence
- Coding is a precondition, not a refinement

Delay

- A new input every 33.333 ms
- Each queue crossed increases elapsed time

Granularity

- A cloud spans ten thousand datagrams
- Stages wait for the last of them

Resource Coupling

- Unrelated duties share one boundary
- Individual tasks cannot be relieved alone



Research Questions and Contributions

1 Introduction

RQ1

Which operations admit a progressive evaluation **in transit**, and which of them yield a value that a later component can consume without recomputing it?

RQ2

What does moving the geometric frontier do to the **load** it withdraws, to the **headroom** it returns, and to the end-to-end latency a viewer perceives?

RQ3

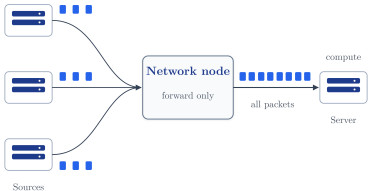
Does the graph sustain 30 fps, and does saturation appear as quantified **backpressure** instead of silent loss?

RQ4

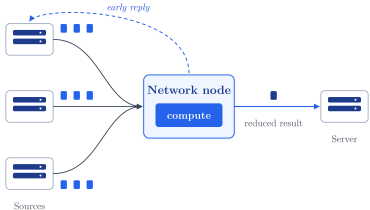
Does the decoded stream keep its **fidelity** once the same arithmetic is resolved elsewhere?

Computation Bound to Packet Traversal

2 Background



Everything is paid at the **destination**



Part of it is settled **on the way**

Origins and Definition

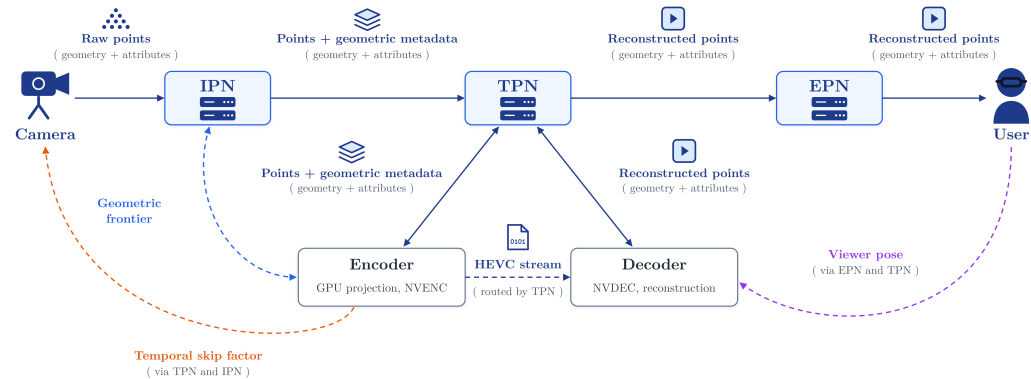
- Chosen because the traffic already crosses it [7]
- Each placement answers for itself [9, 11]

The Data Plane Development Kit

- Kernel bypassed, one isolated core [3]
- Payload resident in **user space** needs no copy

Anatomy of the Redistributed Graph

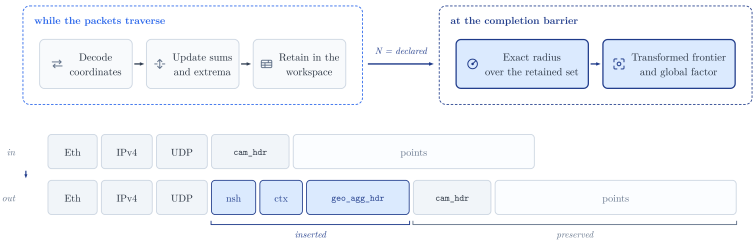
3 Methodology



Service paths and the state they carry follow the chaining architecture [6, 10], while the implementation sources, build recipe and platform specification needed to reproduce the system are publicly available [1].

Geometry Resolved While Forwarding

3 Methodology



Packet-progressive reductions

$$S_N = \sum_{p \in P_N} p$$

$$p_N^{\min} = \min_{p \in P_N} p, \quad p_N^{\max} = \max_{p \in P_N} p$$

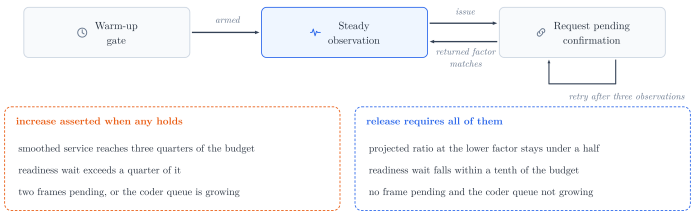
Frame-completing quantities

$$C = \frac{S_N}{N}, \quad r_{\max} = \max_{p \in P} \|p - C\|_2$$

$$f = \frac{0.2\,D}{r_{\max}}, \quad g = (1+\delta)\max\left(\frac{E'_x}{W}, \frac{E'_y}{H}, \frac{E'_z}{W}\right)$$

Regulation Driven by Measured Load

3 Methodology



$$T_{\text{base}} = \frac{1000}{f_{\text{t}}}, \qquad E_n = \alpha\, T_n + (1-\alpha) E_{n-1}, \qquad \rho_n = \frac{E_n}{s\, T_{\text{base}}}$$

α: Smoothing weight, f_{t}: Nominal rate, ρ_n: **Workload ratio**, s: Active factor, T_n: Service interval



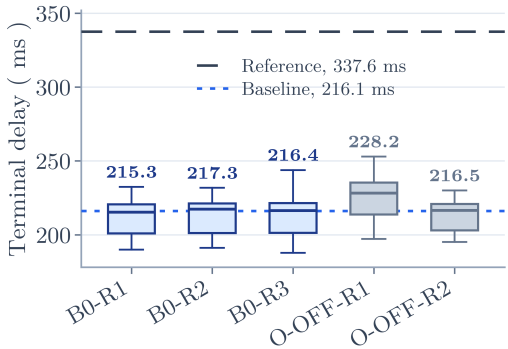
The Conditions Evaluated

4 Performance Evaluation

Family	Runs	Purpose
Baseline	`B0-R1..R3`, with nothing changed	How much the nominal configuration varies on its own, setting the threshold for subsequent comparisons
Ablation	`O-OFF-R1..R2`, both offload switches are disabled	What relocating the geometric frontier costs in practice once everything around it is deliberately held fixed
Packet pressure	`P40`, `P40-NO-T`, `P84`, `M1000`, datagram capacity and grouping	How the pipeline responds as packet pressure rises, with the regulator enabled and then withdrawn

The Repeated Baseline

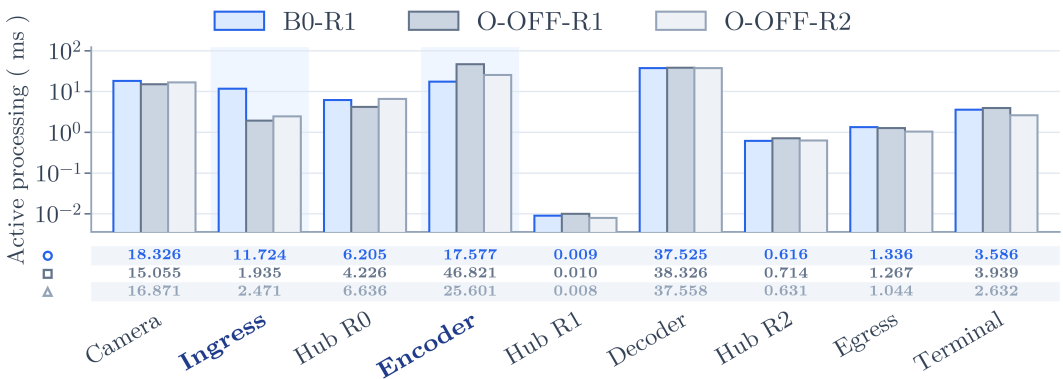
5 Results



The reduction is **35.99 per cent**. Transport, decomposition, control arrangement and deployment all differ besides the site of the frontier, so it belongs to **the redesign as a whole** under these conditions.

Where the Computation Is Paid

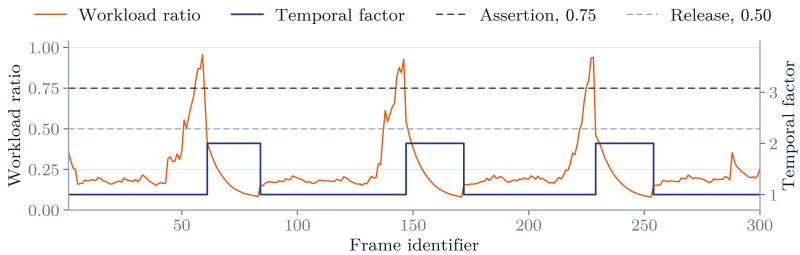
5 Results



	Camera	Ingress	Hub R0	Encoder	Hub R1	Decoder	Hub R2	Egress	Terminal
○	18.326	11.724	6.205	17.577	0.009	37.525	0.616	1.336	3.586
□	15.055	1.935	4.226	46.821	0.010	38.326	0.714	1.267	3.939
△	16.871	2.471	6.636	25.601	0.008	37.558	0.631	1.044	2.632

Pressure Imposed Through the Datagram Rate

5 Results



Halving the points per packet is the only condition that brings the regulator out of its idle state. The cadence settles at **26.074 fps**, so saturation becomes a deliberate adaptation of the source.

What the Redistribution Achieved

6 Conclusions

The redistribution relocates the work without materially changing what reaches the **User**: valid point counts match across all 300 frames, and every loss is attributed to the hop where it occurs.

7
Units following the criterion

35.99 %
Less end-to-end delay

8 / 9
Trials at nominal cadence

Experimental Limitations

- Single host, one sequence, fixed compression
- Core assignment is specific to this topology
- Service representation mirrors the header only

Future Work

- Dynamic workload model beyond static fractions
- Cross the frame barrier with slice segments [13]
- Repeat across other platforms and datasets



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Thank you for listening!

Any questions?



References

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